

ML Foundations — Week 4 Day 1 Summary

UCI Adult (Census Income) Dataset — Predicting Income >\$50K/year | CM-IT Batch 2

Problem Framing

Target: income >50K = positive class (1). **Business objective:** identify likely high-income individuals for targeted outreach (e.g. premium product offers), where each contact has a cost — so minimizing wasted outreach on the wrong people matters more than catching every single high earner. **Base rate:** 23.9% of individuals earn >50K, so accuracy alone is a misleading metric.

Chosen Metric

Primary metric: Precision (supported by F1 and ROC AUC). Precision keeps the contacted list clean — most flagged people should actually be high earners — even if some are missed. Recall was not prioritized because false positives directly cost outreach budget, and the use case tolerates missing some qualified leads.

Baseline Results (Hold-Out Test Set, n=9,769, stratified 80/20 split, random_state=42)

Baseline	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Majority class (always <=50K)	0.761	0.000	0.000	0.000	n/a	n/a
Rule: education_num >= 13	0.747	0.472	0.500	0.486	0.662	0.356

Interpretation: The majority-class baseline reaches 76% accuracy but has zero precision/recall on the positive class — useless for finding high earners. The rule-based baseline (education_num ≥ 13, selected on training data only) trades some raw accuracy for real positive-class performance and wins on precision, recall, F1, and ROC AUC. A real model this week should clear roughly **precision ≥ 0.65–0.70** to be considered a meaningful improvement, since it has access to all 14 features plus learned interactions versus one hand-picked threshold.

Error Analysis & Next Steps

False negatives (missed high earners, n=1,168) skew toward married individuals working long hours without a Bachelor's degree — occupation/experience signal the rule ignores. **False positives** (n=1,308) are often younger, never-married degree-holders early in their careers — a degree alone isn't sufficient. **marital_status** and **hours_per_week** clearly carry information the single-feature rule misses entirely.

Fixes for tomorrow: (1) impute/flag missing workclass, occupation, native_country (~2–6% missing each); (2) group high-cardinality native_country (40+ levels) before encoding; (3) log-transform or binarize the zero-inflated capital_gain/capital_loss; (4) drop or carefully handle fnlwgt (a sampling weight, not a real feature); (5) drop redundant education string column (keep education_num); (6) address 23.9% class imbalance via class weighting / threshold tuning rather than a default 0.5 cutoff.

Primary metric for the week: Precision. The rule baseline sits at ~0.47 precision / 0.50 recall / 0.49 F1. This week's modeling should push precision toward 0.65–0.70 while keeping recall high enough to generate a useful lead volume, tracking F1 and ROC AUC to ensure gains aren't just from an overly conservative threshold.

Full analysis, code, EDA charts, and error samples: ml_foundations_week4_day1.ipynb